

JavaFX MVVM Example

Lab Overview

This lab helps you understand the Model-View-ViewModel (MVVM) design pattern by developing a simple Student Management System using Java and JavaFX.

Example code is available in the lesson PDF file under the name `MvvmJavaFXExample.zip`.

Running the Example Code

- Open Project in VS Code
- Ensure you have JavaFX libraries configured in your project build path.
- `Ctrl + Shift + `` to open the terminal and write `"mvn javafx:run"`.

Adding Lombok to dependency

Lombok is a java library that provides setter and getter methods without declaring them.

- Rename `User` class to `Student` class.
- In `Student` class, remove setter and getter by writing `@Data`, `@NoArgsConstructor`, `@AllArgsConstructor`.

Add Email and Birthdate to Student

In Scenebuilder, I provide 2 more labels which are email and birthdate that allow the user to fill more information.

- In email label, the user can fill their own email address and it will be updated in the listview after the user clicks save.
- In birthdate label, the user can fill their date of birth by clicking the datepicker component in scenebuilder.

Update UI, bindings, and events for the new fields

- In `UserController`, we simply modify the `setupDataBinding()` function so that when the user clicks save, it will store the data in the listview.
- The user can change the data from the listview by double-clicking on their data set, and it will pop up back into the filling label.
- We also modify some functions in `UserViewModel` and `UserController` since they interact with each other related to data binding.

Add Sort Feature

- Add a combobox next to the clear form button and apply a checkbox next to it for ascending data by default and descending data based on user preference.
- The user can sort based on Fullname, email, and date of birth.
- Once the user decides, it will update inside the listview.

Add Search Feature

- Use textfield component as a search feature.
- User can search based on their name, email and dateofbirth.
- The listview will updated based on user search.

Student names of your group members

I have add Team member name by default which is shown when the programming is running

```
public UserViewModel() {  
    users.addAll(  
        new Student("Hongnyheng", "Chin", "hongnyheng@gmail.com",  
            LocalDate.of(2005, 12, 29)),  
        new Student("Vireak", "Rith", "vireakrith@gmail.com",  
            LocalDate.of(2005, 12, 25));  
    }
```